

Experiment 10

Question: Experiment to understand array function in Data science.

Aim

To understand the use of NumPy array functions in data science.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_10():
    # Creation
    a = np.array([10, 20, 30, 40, 50])
    m = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
    print("1D array      :", a)
    print("2D array      :\n", m)
    print("zeros(2,3)      :\n", np.zeros((2, 3)))
    print("ones(2,3)        :\n", np.ones((2, 3)))
    print("arange(0,10,2):", np.arange(0, 10, 2))
    print("linspace(0,1,5):", np.linspace(0, 1, 5))
    print("identity(3)     :\n", np.eye(3))

    # Attributes
    print("\nshape, ndim, size, dtype of m:", m.shape, m.ndim, m.size, m.dtype)

    # Indexing and slicing
    print("\nm[1, 2]          =", m[1, 2])
    print("m[:, 1] (column 2) =", m[:, 1])
    print("m[0:2, 0:2]      :\n", m[0:2, 0:2])
    print("a[a > 25]        =", a[a > 25])          # boolean masking

    # Reshape and transpose
    print("\narange(1,7).reshape(2,3):\n", np.arange(1, 7).reshape(2, 3))
    print("m.T (transpose):\n", m.T)

    # Arithmetic and broadcasting
    print("\na + 5          =", a + 5)
    print("a * 2          =", a * 2)
    print("m + [10,20,30] (broadcasting):\n", m + np.array([10, 20, 30]))

    # Aggregate / statistical functions
    print("\nsum =", a.sum(), "| mean =", a.mean(), "| std =", round(a.std(), 2))
    print("min =", a.min(), "| max =", a.max(), "| argmax =", a.argmax())
    print("column sums of m:", m.sum(axis=0), "| row sums:", m.sum(axis=1))
    print("median =", np.median(a), "| cumsum =", a.cumsum())

    # Linear algebra
    A = np.array([[2, 1], [1, 3]])
    B = np.array([[1, 0], [4, 2]])
    print("\nA @ B (matrix product):\n", A @ B)
    print("det(A) =", round(np.linalg.det(A), 2))
    print("inverse(A):\n", np.linalg.inv(A).round(3))

    # Random numbers
    rng = np.random.default_rng(0)
    print("\nRandom integers :", rng.integers(1, 100, 5))
    print("Random normal    :", rng.normal(0, 1, 3).round(3))
```

exp_10()

Output

```
1D array      : [10 20 30 40 50]
2D array      :
[[1 2 3]
 [4 5 6]
 [7 8 9]]
zeros(2,3)    :
[[0. 0. 0.]
```

```

[0. 0. 0.]]
ones(2,3)      :
[[1. 1. 1.]
 [1. 1. 1.]]
arange(0,10,2): [0 2 4 6 8]
linspace(0,1,5): [0.    0.25 0.5   0.75 1.   ]
identity(3)    :
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]

shape, ndim, size, dtype of m: (3, 3) 2 9 int64

m[1, 2]        = 6
m[:, 1] (column 2) = [2 5 8]
m[0:2, 0:2]    :
[[1 2]
 [4 5]]
a[a > 25]      = [30 40 50]

arange(1,7).reshape(2,3):
[[1 2 3]
 [4 5 6]]
m.T (transpose):
[[1 4 7]
 [2 5 8]
 [3 6 9]]

a + 5   = [15 25 35 45 55]
a * 2   = [ 20 40 60 80 100]
m + [10,20,30] (broadcasting):
[[11 22 33]
 [14 25 36]
 [17 28 39]]

sum = 150 | mean = 30.0 | std = 14.14
min = 10 | max = 50 | argmax = 4
column sums of m: [12 15 18] | row sums: [ 6 15 24]
median = 30.0 | cumsum = [ 10  30  60 100 150]

A @ B (matrix product):
[[ 6  2]
 [13  6]]
det(A) = 5.0
inverse(A):
[[ 0.6 -0.2]
 [-0.2  0.4]]

Random integers : [85 64 51 27 31]
Random normal   : [ 0.105 -0.536  0.362]

```